Yonglin Zhang

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Research Profile

My research lies at the intersection of microscale reaction and transport, interfacial dynamics, and functional materials processing. I investigate how these coupled processes govern the formation of polymer membranes and microcapsules and electroless copper interconnects. I combine experiments, numerical modeling, and scaling analysis to identify governing mechanisms and guide structure control, building on earlier work in solid mechanics and shape-memory composites.

10 journal articles, including 3 first-author and 2 co-first-author papers.

Academic Appointments

The Hong Kong University of Science and Technology **2024 to present**
Postdoctoral Fellow, Department of Mechanical and Aerospace Engineering
Advisor: Prof. Jinglei Yang

Education

PhD in Mechanical Engineering, HKUST **2018 to 2024**
Department of Mechanical and Aerospace Engineering
Advisors: Prof. Jinglei Yang and Prof. Jingshen Wu

MEng in Solid Mechanics, Harbin Institute of Technology **2016 to 2018**
Department of Astronautical Science and Mechanics; Advisor: Prof. Liwu Liu

BEng in Engineering Mechanics, Harbin Institute of Technology **2012 to 2016**
Honors School; Advisor: Prof. Yanju Liu

Research Contributions

Interfacial polymerization and membrane formation

HKUST, Department of Mechanical and Aerospace Engineering | 2024 to present

- Examined reaction–diffusion, water effects, and phase separation in the formation of functional membranes and microcapsules, using experiments and numerical modeling [J1].
- Identified kinetic gating of interfacial instability as a mechanism for regulating membrane topology under flow [J2]. Both this study and the water-effects study [J1] are co-first-author contributions.

Microfluidic copper interconnection and package reliability

HKUST, Center for Engineering Materials and Reliability | 2018 to 2024

- Developed and experimentally validated a simulation framework linking reactive flow and mass transfer to uniform electroless copper deposition [J4].
- Combined experiments, phase-field modeling, and dimensionless scaling to analyze bubble dynamics and microvoid formation in 3D chip interconnection [J3, J4].
- Evaluated electronic package reliability under thermal, mechanical, and electrical loading [C1–C3].

Solid mechanics and adaptive composites

Harbin Institute of Technology, Center for Smart Materials and Structures | 2015 to 2018

- Designed a two-way shape-memory composite combining a shape-memory polymer composite and an electroactive polymer; characterized its thermomechanical response and demonstrated deployable, gripping, and flapping-wing structures [J5].

Journal Articles

Yonglin Zhang is shown in bold. # denotes equal contribution as co-first authors.

Papers as First or Joint First Author

[J1] Liu, B.[#]; **Zhang, Y.[#]**; Zhao, Y.; Prakash, P.; Huai, L.; Ma, H.; Luo, Z.; Goddard, W. A.; Yang, J. (2026). Interfacial Polymerization of TEPA and HMDI: The Role of Water. *ACS Catalysis*, 16(7), 6368-6379. doi:10.1021/acscatal.5c08183

[J2] Zhan, H.[#]; **Zhang, Y.[#]**; Zhao, Y.[#]; Li, D.; Yang, J. (2026). Kinetic gating of interfacial instability directs topology evolution in flow-regulated interfacial polymerization. *Soft Matter*, 22(38), 6464-6474. doi:10.1039/d6sm00601a

[J3] **Zhang, Y.**; Han, Y.; Yao, P.; Guan, Z.; Zhan, H.; Chen, H.; Yang, J. (2026). Bubble dynamics in microfluidic electroless copper interconnection for advanced 3D chip integration. *International Communications in Heat and Mass Transfer*, 172, 110454. doi:10.1016/j.icheatmasstransfer.2025.110454

[J4] **Zhang, Y.**; Yao, P.; Han, Y.; Yang, J.; Chen, H.; Wu, J.; Yang, J. (2022). Microfluidic electroless deposition for uniform stacking chip interconnection: Simulation framework and experimental validation. *Chemical Engineering Journal*, 434, 134684. doi:10.1016/j.cej.2022.134684

[J5] **Zhang, Y.**; Liu, T.; Lan, X.; Liu, Y.; Leng, J.; Liu, L. (2022). A compliant robotic grip structure based on shape memory polymer composite. *Composites Communications*, 36, 101383. doi:10.1016/j.coco.2022.101383

Additional Journal Articles

[J6] Han, Y.; Du, W.; **Zhang, Y.**; Qiu, C.; Law, M.; Zhao, Y.; Tang, B. Z.; Wang, Y.; Yang, J. (2026). Programming Interfacial Polymerization: Machine Learning Unveils Quantitative Rational Design Rules for Microcapsules and Beyond. *Advanced Materials*, 38(12), e17708. doi:10.1002/adma.202517708

[J7] Yu, S.; Mu, Y.; Gu, H.; **Zhang, Y.**; Zhao, S.; Liang, H.; Hu, B.; Sun, J.; Chen, X.; Zeng, L.; Bai, J.; Yang, J. (2025). Achieving fast Zn dynamics in high-mass-loading cathode via 3D-printed hierarchical graphene architectures in zinc-ion batteries. *Chemical Engineering Journal*, 526, 170624. doi:10.1016/j.cej.2025.170624

[J8] Liu, T.; Lin, C.; **Zhang, Y.**; Cai, J.; Yang, J. (2025). Viscoelastic negative stiffness metamaterial with multistage load bearing and programmable energy absorption ability. *International Journal of Smart and Nano Materials*, 16(1), 1-23. doi:10.1080/19475411.2024.2428172

[J9] Yu, S.; **Zhang, Y.**; Mu, Y.; Guo, B.; Zeng, G.; Xiang, Y.; Zeng, L.; Bai, J.; Yang, J. (2024). Towards reusable 3D-printed graphite framework for zinc anode in aqueous zinc battery. *Energy Storage Materials*, 70, 103454. doi:10.1016/j.ensm.2024.103454

[J10] Liu, T.; Deng, R.; **Zhang, Y.**; Yang, J.; Cai, J. (2024). Composite negative stiffness structure with tunable and temperature-dependent properties induced by viscoelastic and shape-memory materials. *Composites Communications*, 48, 101937. doi:10.1016/j.coco.2024.101937

Conference Papers

[C1] Yao, P.; Yang, J.; **Zhang, Y.**; Fan, X.; Chen, H.; Yang, J.; Wu, J. (2022). Physics-based Nested-ANN Approach for Fan-Out Wafer Level Package Reliability Prediction. 72nd IEEE Electronic Components and Technology Conference (ECTC). IEEE.

[C2] Zhang, Y.; Chen, H.; Fan, H.; Yang, J.; Wu, J. (2021). Numerical Investigation on Microfluidic Electroless Deposition for Uniform Copper Pillar Microbumps Interconnection. 71st IEEE Electronic Components and Technology Conference (ECTC), pp. 402–407. IEEE.

[C3] Zhang, Y.; Chen, H.; Xue, K.; Yang, J.; Wu, J. (2020). The Finite Element Model of the Effect of the Interface Behavior of Corner Bond on the Reliability of the BGA Package under Thermal Cycling. 21st International Conference on Electronic Packaging Technology (ICEPT), pp. 1–5. IEEE.

Patent Publication

[P1] Liu, Y.; Leng, J.; **Zhang, Y.**; Liu, T. An Iris Identification System to Stimulate the Deformation of Shape Memory Polymer. Chinese patent application publication CN105678139A, 15 June 2016.

Invited Talks and Conference Presentations

2026 | Oral presentation. Multiscale Mechanisms and Structure Regulation of Interfacial Polymerization via Solvent Engineering for Functional Membrane. E-MRS Spring Meeting, Strasbourg, France.

2025 | Invited talk. Multiscale Interaction Mechanisms and Structural Regulation of Interfacial Polymerization Based on Solvent Engineering. 2nd Chinese Conference on Smart Materials and Structural Systems (SMSS), Guangzhou, China.

2024 | Conference presentation. Microscale Transport Phenomena and Bubble Dynamics in Microfluidic Electroless Copper Interconnection for Chip 3D Integration. ASME InterPACK, San Jose, USA.

2021 | Conference presentation. Numerical Investigation on Microfluidic Electroless Deposition for Uniform Copper Pillar Microbumps Interconnection. 71st IEEE ECTC, virtual.

Invited talk. Symmetry-Breaking Guided Formation and Regulation of Interfacial Polymerization for Functional Membrane Green Manufacturing: In-situ Monitoring and Modeling. 32nd International Conference on Computational & Experimental Engineering and Sciences, Hong Kong.

Research Support

Research Talent Hub, Innovation and Technology Commission, Hong Kong	2025
UGC Research Travel Grant	2024 and 2026

Honors and Awards

The Bright Dream Robotics Scholarship	2023
Special Scholarship	2016
People Scholarship	2013 to 2016
Honorable Prize, Mathematical Contest in Modeling	2014

Academic Service

Journal reviewer: Journal of Electronic Packaging; Composites Part A: Applied Science and Manufacturing.

Professional memberships: IEEE; ASME.